Name _____ Class _____

Higher Physics

Assignment: mark it yourself — worksheet



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Remember

This pack is for practice **before** you write your report. It is never used while you write it.

Your tick list is the official **Instructions for candidates** in the Higher Physics coursework assessment task. They are the only tick list you may use in the report stage.

Use this worksheet with the **reader** on your iPad. The reader has the five scripts, the tick lists and the answers.

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Grid 1: script 1, Planck's constant from LEDs

Script 1 is part 3 of the reader (7 pages). For each section, **write** your mark and what the marker would look for: is it here, or missing?

Item	Section	Max	Your mark	What is here, or what is missing?
1	Aim	1		
2	Underlying physics	3		
3a	Method summary	1		
3b	Raw data	1		
3c	Data table	1		
3d	Second data	1		
3e	Citation and reference	1		
4a	Scales	1		
4b	Labels and units	1		
4c	Plotting and line	1		
5	Uncertainties	2		
6	Analysis	1		
7	Conclusion	1		
8	Evaluation	3		
9	Title and structure	1		
	Total	**20**		

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Grid 2: script 2, Internal resistance

Script 2 is part 4 of the reader (9 pages). For each section, **write** your mark and what the marker would look for: is it here, or missing?

Item	Section	Max	Your mark	What is here, or what is missing?
1	Aim	1		
2	Underlying physics	3		
3a	Method summary	1		
3b	Raw data	1		
3c	Data table	1		
3d	Second data	1		
3e	Citation and reference	1		
4a	Scales	1		
4b	Labels and units	1		
4c	Plotting and line	1		
5	Uncertainties	2		
6	Analysis	1		
7	Conclusion	1		
8	Evaluation	3		
9	Title and structure	1		
	Total	**20**		

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Grid 3: script 3, Peak a.c. and d.c. voltage

Script 3 is part 5 of the reader (5 pages). For each section, **write** your mark and what the marker would look for: is it here, or missing?

Item	Section	Max	Your mark	What is here, or what is missing?
1	Aim	1		
2	Underlying physics	3		
3a	Method summary	1		
3b	Raw data	1		
3c	Data table	1		
3d	Second data	1		
3e	Citation and reference	1		
4a	Scales	1		
4b	Labels and units	1		
4c	Plotting and line	1		
5	Uncertainties	2		
6	Analysis	1		
7	Conclusion	1		
8	Evaluation	3		
9	Title and structure	1		
	Total	**20**		

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Grid 4: script 4, Irradiance and distance

Script 4 is part 6 of the reader (10 pages). For each section, **write** your mark and what the marker would look for: is it here, or missing?

Item	Section	Max	Your mark	What is here, or what is missing?
1	Aim	1		
2	Underlying physics	3		
3a	Method summary	1		
3b	Raw data	1		
3c	Data table	1		
3d	Second data	1		
3e	Citation and reference	1		
4a	Scales	1		
4b	Labels and units	1		
4c	Plotting and line	1		
5	Uncertainties	2		
6	Analysis	1		
7	Conclusion	1		
8	Evaluation	3		
9	Title and structure	1		
	Total	**20**		

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Grid 5: script 5, Refractive index of water

Script 5 is part 7 of the reader (6 pages). For each section, **write** your mark and what the marker would look for: is it here, or missing?

Item	Section	Max	Your mark	What is here, or what is missing?
1	Aim	1		
2	Underlying physics	3		
3a	Method summary	1		
3b	Raw data	1		
3c	Data table	1		
3d	Second data	1		
3e	Citation and reference	1		
4a	Scales	1		
4b	Labels and units	1		
4c	Plotting and line	1		
5	Uncertainties	2		
6	Analysis	1		
7	Conclusion	1		
8	Evaluation	3		
9	Title and structure	1		
	Total	**20**		

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Focus tasks

The tasks are in order of marks lost, biggest first. Each one says why. Script page numbers are the ones at the top of each page in the reader.

Task 1: Sorting evaluation statements (8)

Why: More marks are lost here than in any other section. The 2025 course report says "Few candidates were able to give more than one evaluative statement with appropriate justification."

Look at: script 4, page 9; script 5, page 6.

1 **Tick** whether each statement scores a mark.

	Scores	No mark
Script 4, statement 2: more repeats would improve accuracy		
Script 4, statement 3: measure the background light and subtract it		
Script 4, statement 4: use a ruler with 1 mm divisions		
Script 5, statement 1: check the water is at room temperature		
Script 5, statement 2: use a semicircular box		
Script 5, statement 3: more repeats, due to the variation in the results		

2 **Name** the evaluation tick-list item that each scoring statement uses.

Background light:

Semicircular box:

More repeats:

3 **Rewrite** script 5's temperature statement so that it would gain a mark.

The temperature of the water

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Task 2: Topic level and the underlying physics (2)

Why: The second biggest loss of marks. The 2025 course report says pupils with National 5 topics "often struggled to include any physics that was at Higher level".

Look at: script 2, pages 1 and 2; script 3, page 1.

1 Tick each item from the underlying physics tick list that each script includes.

	Script 2	Script 3
relationships or equations		
definitions of symbols used		
explanations or justifications of relationships or equations		
explanations of physical properties		
copies of diagrams that you would find difficult to draw		

2 Explain why script 3 scored 0 out of 3 for its underlying physics.

Script 3 scored 0 because

3 Suggest Higher physics that script 3 could have described for its aim.

Script 3 could have described

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Task 3: The conclusion states the value (7)

Why: One of the biggest losses of marks. In the SQA scripts, a conclusion that gives no value for the aim scores 0.

Look at: script 1, pages 1 and 5; script 5, page 6.

1 State the aim of script 1.

The aim of script 1 is

2 Explain why script 1's conclusion scored 0, when script 5's scored 1.

Script 1's conclusion scored 0 because

3 Write the first sentence of a better conclusion for script 1.

My experiment gave a value of Planck's constant of

Task 4: Proportional or linear? (6, 7)

Why: Analysis and conclusion marks are often lost here. The 2025 course report says to use "the terminology 'linear relationship', as that can be applied to any best-fit straight line, whether it passes through the origin or not".

Look at: script 4, pages 5, 7 and 9.

1 State whether the line on script 4's graph passes through the origin.

The line

2 Explain the difference between a linear relationship and direct proportion.

Both have a straight line, but direct proportion

3 Write a valid conclusion for script 4.

My results show

Task 5: A citation is not a URL (3e)

Why: A common lost mark. The 2025 course report says "using a full URL as the citation" is "never appropriate".
Look at: script 1, pages 4 and 7; script 2, pages 1 and 9; script 3, pages 4 and 5; script 4, pages 8 and 10.

1 Tick each thing that is true for each script. Columns 1 to 4 are the scripts.

	1	2	3	4
a number or label sits beside the data				
the same number sits beside the reference				
the reference is complete				

2 Explain why script 1 did not gain the citation mark.

Script 1 did not gain the mark because

3 Write a citation and a reference that would gain the mark for a website.

Beside the data:

In the reference list:

Task 6: A method summary (3a)

Why: A common lost mark. The 2025 course report says many pupils "did not summarise the method and instead gave a detailed, often step-by-step, description".

Look at: script 1, page 1; script 3, page 1; script 4, page 3.

1 Tick each thing that is true for each method summary.

	Script 1	Script 3	Script 4
says what was changed			
says what was measured			
names the measuring instruments			

2 Write a method summary for script 4's experiment in two sentences.

In this experiment,

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